```sql
USE oct_batch;

-- Retrieve all students ordered by enroll_date
SELECT *
FROM students
ORDER BY enroll_date ASC;

-- Retrieve all courses ordered by credits (highest first)
SELECT *
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	first_name	last_name	email	phone	enroll_date
1	Aarav	Sharma	aarav.sharma@email.com	9876500001	2023-01-15
2	Isha	Patel	isha.patel@email.com	9876500002	2023-02-10
3	Dev	Menon	dev.menon@email.com	9876500003	2023-03-05
4	Ananya	Rao	ananya.rao@email.com	9876500004	2023-04-20
5	Rohan	Kapoor	rohan.kapoor@email.com	9876500005	2023-05-12
6	Meera	Joshi	meera.joshi@email.com	9876500006	2023-06-01
7	Aditya	Verma	aditya.verma@email.com	9876500007	2023-06-15
8	Kavya	Nair	kavya.nair@email.com	9876500008	2023-07-02
9	Siddharth	Gupta	siddharth.gupta@email.com	9876500009	2023-07-18
10	Neha	Bose	neha.bose@email.com	9876500010	2023-08-05

```sql
-- Retrieve all courses ordered by credits (highest first)
SELECT *
FROM courses
ORDER BY credits DESC;

-- Show all enrollments with student name + course name + grade
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_id	course_name	instructor	department	credits
101	Database Systems	Dr. Mehta	Computer Science	4
124	Electromagnetism	Dr. Singh	Physics	4
129	Thermodynamics	Dr. Das	Physics	4
104	Modern Physics	Dr. Singh	Physics	4
136	Cloud Computing	Dr. Verma	Computer Science	4
106	Operating Systems	Dr. Kumar	Computer Science	4
150	Artificial Intelligence Applications	Dr. Das	Computer Science	4
119	Nuclear Physics	Dr. Das	Physics	4
109	Quantum Mechanics	Dr. Das	Physics	4
134	Optics	Dr. Iyer	Physics	4

```sql
-- Show all enrollments with student name + course name + grade
SELECT e.enrollment_id, s.first_name, s.last_name, c.course_name, e.grade
FROM enrollments_1 as e
JOIN students s ON e.student_id = s.student_id
JOIN courses c ON e.course_id = c.course_id
ORDER BY e.enroll_date ASC;

-- Count number of students enrolled in each course
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

enrollment_id	first_name	last_name	course_name	grade
1	Aarav	Sharma	Database Systems	A
2	Isha	Patel	Linear Algebra	B+
3	Dev	Menon	Microeconomics	A-
4	Ananya	Rao	Modern Physics	B
5	Rohan	Kapoor	Creative Writing	A
6	Meera	Joshi	Operating Systems	B+
7	Aditya	Verma	Statistics	A
8	Kavya	Nair	Macroeconomics	B-
9	Siddharth	Gupta	Quantum Mechanics	A-
10	Neha	Bose	British Literature	B+

```sql
-- Count number of students enrolled in each course
SELECT c.course_name, COUNT(*) AS student_count
FROM enrollments_1 as e
JOIN courses c ON e.course_id = c.course_id
GROUP BY c.course_name
ORDER BY student_count DESC;

-- Average GPA per course
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_name	student_count
Database Systems	1
Linear Algebra	1
Microeconomics	1
Modern Physics	1
Creative Writing	1
Operating Systems	1
Statistics	1
Macroeconomics	1
Quantum Mechanics	1
British Literature	1

```sql
-- Average GPA per course
SELECT c.course_name,
       ROUND(AVG(CASE e.grade
           WHEN 'A' THEN 4.0
           WHEN 'A-' THEN 3.7
           WHEN 'B+' THEN 3.3
           WHEN 'B' THEN 3.0
           WHEN 'B-' THEN 2.7
       END),2) AS avg_gpa
FROM enrollments_1 as e
JOIN courses c ON e.course_id = c.course_id
GROUP BY c.course_name
ORDER BY avg_gpa DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content

course_name	avg_gpa
Database Systems	4.00
Nuclear Physics	4.00
Modern Drama	4.00
Creative Writing	4.00
Probability	4.00
Statistics	4.00
Data Structures	4.00
Artificial Intelligence	4.00
Poetry Analysis	4.00
Game Theory	4.00

```sql
-- Count enrollments by status
SELECT status, COUNT(*) AS enrollment_count
FROM enrollments_1
GROUP BY status;

-- Find students who have enrolled in more than 2
```

Result Grid | Filter Rows: | Export: | Wrap Ce

	status	enrollment_count
▶	enrolled	50

```sql
-- Find courses with more enrollments than the average
SELECT c.course_name
FROM courses c
WHERE c.course_id IN (
    SELECT course_id
    FROM enrollments_1
    GROUP BY course_id
    HAVING COUNT(*) > (
        SELECT AVG(enrollment_count)
        FROM (
            SELECT COUNT(*) AS enrollment_count
            FROM enrollments_1
            GROUP BY course_id
```

Grid | Filter Rows: | Export: | Wrap Cell Content:

urse_name

```sql
SELECT s.first_name, s.last_name, c.course_name, e.grade
FROM enrollments_1 e
JOIN students s ON e.student_id = s.student_id
JOIN courses c ON e.course_id = c.course_id
WHERE (
    CASE e.grade
        WHEN 'A' THEN 4.0
        WHEN 'A-' THEN 3.7
        WHEN 'B+' THEN 3.3
        WHEN 'B' THEN 3.0
        WHEN 'B-' THEN 2.7
    END
) > (
    SELECT AVG(
        CASE grade
            WHEN 'A' THEN 4.0
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	course_name	grade

```
        WHEN 'B' THEN 3.0
        WHEN 'B-' THEN 2.7
    END
) > (
    SELECT AVG(
        CASE grade
            WHEN 'A' THEN 4.0
            WHEN 'A-' THEN 3.7
            WHEN 'B+' THEN 3.3
            WHEN 'B' THEN 3.0
            WHEN 'B-' THEN 2.7
        END
    )
    FROM enrollments_1
    WHERE course_id = e.course_id
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	first_name	last_name	course_name	grade

```sql
-- Rank courses by number of enrollments
SELECT c.course_name, COUNT(e.enrollment_id) AS times_enrolled,
       RANK() OVER (ORDER BY COUNT(e.enrollment_id) DESC) AS course_rank
FROM enrollments_1 e
JOIN courses c ON e.course_id = c.course_id
GROUP BY c.course_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_name	times_enrolled	course_rank
Database Systems	1	1
Linear Algebra	1	1
Microeconomics	1	1
Modern Physics	1	1
Creative Writing	1	1
Operating Systems	1	1
Statistics	1	1
Macroeconomics	1	1
Quantum Mechanics	1	1

```sql
-- Rank students by number of courses taken
SELECT s.first_name, s.last_name, COUNT(e.enrollment_id) AS course_count,
       RANK() OVER (ORDER BY COUNT(e.enrollment_id) DESC) AS student_rank
FROM enrollments_1 e
JOIN students s ON e.student_id = s.student_id
GROUP BY s.student_id, s.first_name, s.last_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	course_count	student_rank
Aarav	Sharma	1	1
Isha	Patel	1	1
Dev	Menon	1	1
Ananya	Rao	1	1
Rohan	Kapoor	1	1
Meera	Joshi	1	1
Aditya	Verma	1	1
Kavya	Nair	1	1
Siddharth	Gupta	1	1

```sql
-- Use ROW_NUMBER() to assign sequence of enrollments per student
SELECT s.first_name, s.last_name, e.enrollment_id, e.enroll_date,
       ROW_NUMBER() OVER (PARTITION BY s.student_id ORDER BY e.enroll_date) AS enrollment_number
FROM enrollments_1 e
JOIN students s ON e.student_id = s.student_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	enrollment_id	enroll_date	enrollment_number
Aarav	Sharma	1	2023-01-20	1
Isha	Patel	2	2023-02-15	1
Dev	Menon	3	2023-03-10	1
Ananya	Rao	4	2023-04-25	1
Rohan	Kapoor	5	2023-05-15	1
Meera	Joshi	6	2023-06-05	1
Aditya	Verma	7	2023-06-20	1
Kavya	Nair	8	2023-07-10	1
Siddharth	Gupta	9	2023-07-25	1

```sql
-- Use a CTE to find students with more than 3 enrollments
WITH student_enroll_count AS (
    SELECT student_id, COUNT(*) AS enroll_count
    FROM enrollments_1
    GROUP BY student_id
)
SELECT s.first_name, s.last_name, se.enroll_count
FROM student_enroll_count se
JOIN students s ON se.student_id = s.student_id
WHERE se.enroll_count > 3
ORDER BY se.enroll_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	enroll_count